AGENTRX · TAU-BENCH RETAIL · RUN TAU_EVAL

AgentRx Tau-bench Evaluation

We ran the full AgentRx diagnosis pipeline on all 29 failed Tau-bench retail trajectories and scored the LLM judge against the human ground-truth labels. The judge found the right failure category in 7 of 29 tasks and the exact critical step in 7 of 29.

Run by **Pranudeep Metuku** Dates **Oct 12–13, 2026** Code **microsoft/AgentRx** at `7a18c79` LLM backend **GitHub Copilot CLI**

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7 / 29	7 / 29	55%	3 / 29
Failure category correct (24%)	Exact critical step correct (24%)	Step within ±3 of ground truth	Both category and exact step correct

SETUP

What we ran

ITEM	VALUE
Input	`data/tau_retail/tau_dataset_failed.json`: 29 failed retail customer-service conversations (all `reward = 0`), 20–62 messages each, 1,064 steps in total
Ground truth	`data/ground_truth/tau_ground_truth.json`: one labelled critical step and failure category per task
Domain	`tau` (retail policy and tool schemas from the repo)
Models	Static and dynamic invariants: Copilot `auto`, which routed between gpt-6-luna, gemini-3.8-flash, claude-sonnet-5.5 and gpt-6.1-sol. Judge: `gpt-6-luna` pinned.
Judge mode	`combined` prompt (taxonomy + checklist + few-shot), violations supplied after the trajectory
Environment	macOS, Python 3.13.5, `pip install -e .` into a local venv

```
python run.py data/tau_retail/tau_dataset_failed.json \
  --domain tau --endpoint copilot \
  --ground-truth data/ground_truth/tau_ground_truth.json \
  --run-name tau_eval
```

PIPELINE

Stages and timeline

#	STAGE	WHAT IT DID	WHEN	STATUS
1	IR	Converted the 29 raw conversations into the canonical trajectory format	Oct 12 18:41	done
2	Static invariants	Generated policy, tool and structure rules from the retail policy	Oct 12 18:42	done

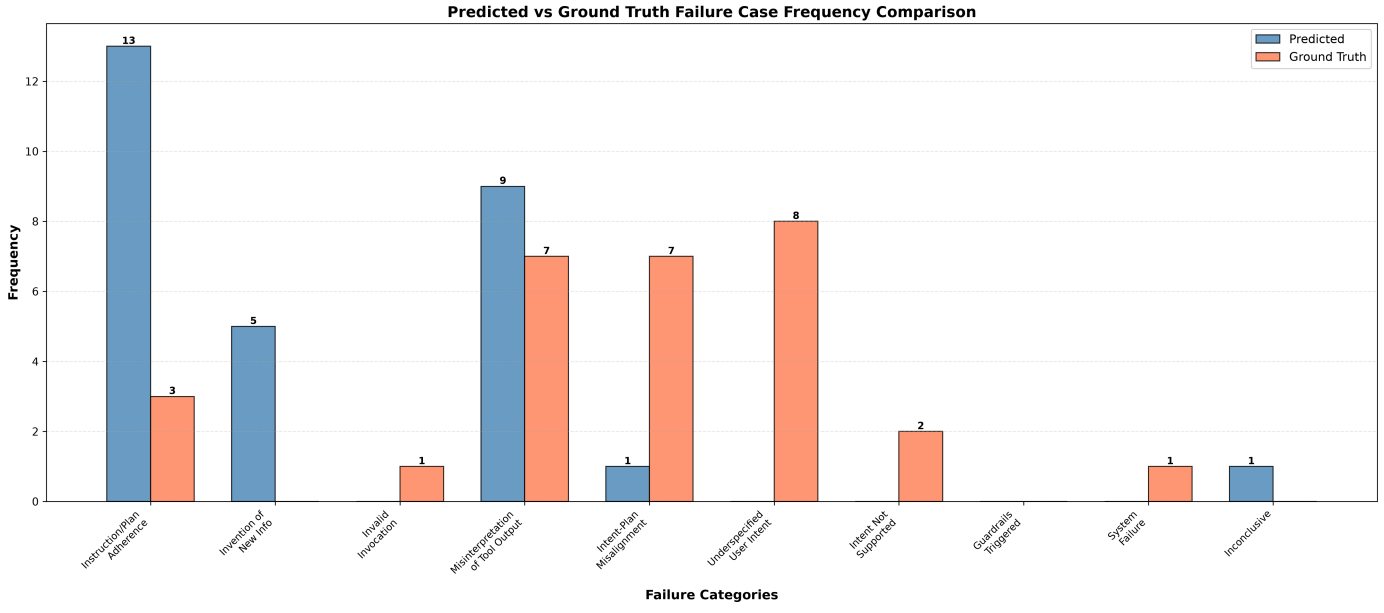
#	STAGE	WHAT IT DID	WHEN	STATUS
3	Dynamic invariants	One LLM call per step (1,064 steps) to write step-specific checks. Interrupted after about 2 tasks, then restarted from this stage.	Oct 12 19:00 – 23:31	done
4	Check	Ran 325 Python invariant checks; recorded 41 violations across 17 tasks	Oct 12 23:31	done
5	Judge	LLM picked the critical step and failure category for each task, then compared with ground truth. A first attempt was cancelled; rerun with `--from-stage judge` on `gpt-6-luna`.	Oct 13 18:25 – 18:33	done
6	Report	Plotted predicted vs ground-truth category frequencies	Oct 13 18:33	done

Issues we hit along the way

- **Default model not available.** The repo's default Copilot model, `claude-opus-4.6`, is rejected by our Copilot plan. We switched to `auto` to get the run going.
- **`auto` was expensive.** It routed some calls to models costing 7–11 credits each. Copilot's local usage records show about 2,150 AI credits in total, nearly all spent on dynamic invariant generation. Pinning `gpt-6-luna` for the judge brought that stage down to about 3 credits for 29 tasks.
- **Resuming the dynamic stage restarts it.** `--from-stage dynamic` regenerates every task; it does not skip ones already finished.
- **Few-shot examples are missing from the repo.** The judge looks for `agentrx/judge/few_shot_examples/`, which isn't checked in. It logged the misses and judged without examples.
- **27% of invariant checks crashed.** 89 of 325 generated checks failed in the checker sandbox and were skipped (details below).

RESULTS

Failure category



Pipeline output `plots/comparison.png`. Blue: judge predictions. Orange: ground truth. The judge overpredicts Instruction/Plan Adherence and almost never predicts the intent categories.

GROUND-TRUTH CATEGORY	TASKS	CORRECT	WHAT THE JUDGE SAID INSTEAD
Misinterpretation of Tool Output	7	6	Instruction/Plan Adherence (1)

GROUND-TRUTH CATEGORY	TASKS	CORRECT	WHAT THE JUDGE SAID INSTEAD
Underspecified User Intent	8	0	Instruction/Plan Adherence (4), Invention (2), Misinterpretation (1), Inconclusive (1)
Intent-Plan Misalignment	7	0	Instruction/Plan Adherence (5), Invention (1), Misinterpretation (1)
Instruction/Plan Adherence	3	1	Invention (1), Misinterpretation (1)
Intent Not Supported	2	0	Instruction/Plan Adherence (2)
Invalid Invocation	1	0	Intent-Plan Misalignment (1)
System Failure	1	0	Invention (1)
Total	**29**	**7**	

Critical step

The judge's step guess is usually close but often slightly early: 17 guesses were before the true step, 7 exact and 5 after. The median error is −1 step. When the category was right, the step was off by 1.3 on average; when it was wrong, by 8.2.

TOLERANCE	TASKS	ACCURACY
Exact	7	24%
Within ±1	11	38%
Within ±2	14	48%
Within ±3	16	55%
Within ±5	20	69%

Invariant evidence

The checker found a violation at or next to the true failure step in only 7 of 29 tasks. When it did, the judge's category accuracy more than doubled. Violations elsewhere in the trajectory tended to pull the step guess away from the real failure.

GROUP	TASKS	CATEGORY CORRECT	STEP WITHIN ±3
A violation within ±1 step of the true failure	7	57%	71%
No violation near the true failure	22	14%	50%
Any violations	17	24%	47%
No violations at all	12	25%	67%

Why 89 checks crashed

CHECKS	ERROR IN GENERATED CHECK CODE
56	`Import of 'decimal' is not allowed in invariant checks`
16	`name 'next' is not defined` (built-in blocked by the sandbox)
11	`Expecting value: line 1 column 1` (empty or invalid JSON)
4	`name 'eval' is not defined`
1	`Import of 'ast' is not allowed`

CHECKS	ERROR IN GENERATED CHECK CODE
1	`unhashable type: 'dict'`

Per-task results

Step numbers are as recorded by the judge and the ground truth. “Crashed” counts checks that failed to run, out of all checks for that task.

TASK	STEPS	GT STEP	PRED STEP	DIFF	GROUND-TRUTH CATEGORY	PREDICTED CATEGORY	CATEGORY	VIOLATIONS	CRASHED
2	32	7	6	-1	Misinterpretation of Tool Output	Misinterpretation of Tool Output	correct	1	1/14
3	34	7	7	0	Misinterpretation of Tool Output	Misinterpretation of Tool Output	correct	1	3/7
4	48	15	14	-1	Misinterpretation of Tool Output	Misinterpretation of Tool Output	correct	16	6/37
12	26	19	15	-4	Instruction/Plan Adherence	Invention of New Info	wrong	0	0/0
20	62	21	61	+40	Misinterpretation of Tool Output	Instruction/Plan Adherence	wrong	2	9/19
21	48	28	41	+13	Underspecified User Intent	Instruction/Plan Adherence	wrong	1	5/10
28	42	33	31	-2	Intent-Plan Misalignment	Instruction/Plan Adherence	wrong	1	11/12
31	42	31	29	-2	Intent-Plan Misalignment	Misinterpretation of Tool Output	wrong	0	8/34
32	42	11	11	0	Misinterpretation of Tool Output	Misinterpretation of Tool Output	correct	0	0/10
34	30	17	13	-4	Invalid Invocation	Intent-Plan Misalignment	wrong	1	0/2
38	46	37	37	0	Misinterpretation of Tool Output	Misinterpretation of Tool Output	correct	0	6/10
39	26	17	17	0	Intent-Plan Misalignment	Invention of New Info	wrong	0	0/2
41	24	17	17	0	Intent-Plan Misalignment	Instruction/Plan Adherence	wrong	1	1/8
47	20	19	8	-11	Intent Not Supported	Instruction/Plan Adherence	wrong	1	1/2
55	38	32	27	-5	Underspecified User Intent	Instruction/Plan Adherence	wrong	1	8/10
59	26	14	11	-3	Underspecified User Intent	Invention of New Info	wrong	0	0/6
63	36	32	28	-4	Underspecified User Intent	Misinterpretation of Tool Output	wrong	1	0/2
71	36	24	27	+3	Underspecified User Intent	Instruction/Plan Adherence	wrong	5	0/25
72	28	21	21	0	Intent-Plan Misalignment	Instruction/Plan Adherence	wrong	0	2/10

TASK	STEPS	GT STEP	PRED STEP	DIFF	GROUND-TRUTH CATEGORY	PREDICTED CATEGORY	CATEGORY	VIOLATIONS	CRASHED
74	30	21	27	+6	Instruction/Plan Adherence	Instruction/Plan Adherence	correct	0	2/5
79	26	20	0	-20	Underspecified User Intent	Inconclusive	wrong	0	0/10
80	26	26	25	-1	System Failure	Invention of New Info	wrong	0	4/4
82	28	20	27	+7	Underspecified User Intent	Invention of New Info	wrong	0	0/10
87	38	37	20	-17	Intent-Plan Misalignment	Instruction/Plan Adherence	wrong	2	0/18
91	40	34	21	-13	Underspecified User Intent	Instruction/Plan Adherence	wrong	1	1/2
98	38	21	21	0	Intent-Plan Misalignment	Instruction/Plan Adherence	wrong	1	0/4
99	46	35	33	-2	Instruction/Plan Adherence	Misinterpretation of Tool Output	wrong	0	8/12
104	58	57	56	-1	Misinterpretation of Tool Output	Misinterpretation of Tool Output	correct	2	7/24
105	48	43	14	-29	Intent Not Supported	Instruction/Plan Adherence	wrong	3	6/16

DISCUSSION

Key findings

Strong on tool-output errors, weak on intent errors.

6 of 7 Misinterpretation of Tool Output cases were correct. 0 of 17 intent-related cases (Underspecified, Misalignment, Not Supported) were correct.

Instruction/Plan Adherence is the judge's fallback answer.

It was predicted 13 times; only 3 tasks actually have that label.

Step guesses are close but early.

Accuracy rises from 24% exact to 69% within ±5. Five tasks are off by exactly −1, so a step-numbering offset between the IR and the ground truth may explain part of the gap.

Evidence quality drives accuracy.

With a violation near the true step, category accuracy was 57%; without one, 14%. The checker caught the real failure in only 7 tasks, and 27% of its checks crashed.

> These numbers are not directly comparable to the AgentRx paper. This run had no few-shot examples, used a mix of models for invariant generation, used a low-cost judge model, and is a single judge pass on 29 tasks, where one task moves accuracy by 3.4 points.

Next steps

1. **Check the step-numbering offset.** Compare checker `step_index` with ground-truth `step_number` on tasks 2, 3, 4 and 104. No LLM cost.
2. **Re-judge with a stronger model.** About 29 calls with `--from-stage judge`. Separates judge quality from evidence quality.
3. **Add few-shot examples**, starting with Intent-Plan Misalignment, Underspecified User Intent and Intent Not Supported.
4. **Fix the checker sandbox** (allow `decimal` and `next`, retry invalid JSON), then rerun the check and judge stages.
5. **Pin a low-cost model for any rerun of dynamic invariants** instead of `auto`, and budget roughly one call per trajectory step.

Reproduce

```
git clone https://github.com/pranudeepmetuku10/AgentRx.git && cd AgentRx
python3 -m venv .venv && source .venv/bin/activate
pip install -e .
cp .env.example .env
# in .env: AGENT_VERIFY_ENDPOINT_TYPE=copilot
#          AGENT_VERIFY_COPILOT_MODEL=gpt-6-luna
# requires the Copilot CLI on PATH and gh auth login

python run.py data/tau_retail/tau_dataset_failed.json \
  --domain tau --endpoint copilot \
  --ground-truth data/ground_truth/tau_ground_truth.json \
  --run-name tau_eval

# resume a stopped run from any stage
python run.py data/tau_retail/tau_dataset_failed.json --domain tau \
  --ground-truth data/ground_truth/tau_ground_truth.json \
  --run-dir runs/tau_eval --from-stage judge
```

Saved outputs (judge predictions, checker violations, plots) are in `results/tau_eval/` of pranudeepmetuku10/AgentRx.

Source: `runs/tau_eval/judge_output/runs/run1.json`, `runs/tau_eval/checker_results/` and Copilot CLI usage records. Paper: AgentRx: Diagnosing AI Agent Failures from Execution Trajectories.